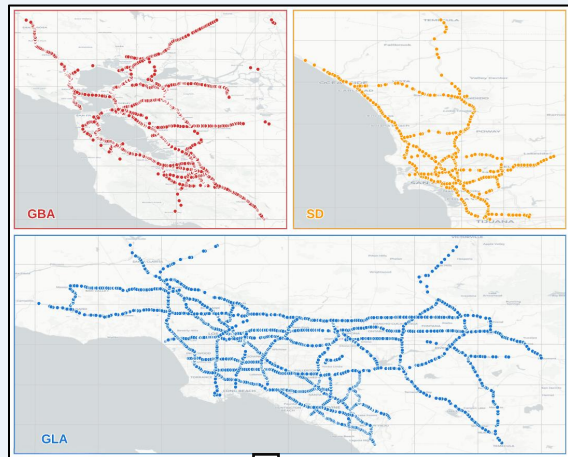


Spatio-temporal Data

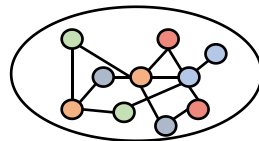


Visual Modality Transformation

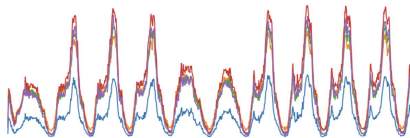
Multi-modal Representations of Spatio-temporal Data



Textual



Structural



Multivariate Time Series

Existing Methods



Spatial Learning



Temporal Learning

Computational cost increases for large-scale spatio-temporal data!

Correlation View

Interaction  Structure
temporal-weighted
correlation adjacency

Temporal View

Dynamic  Cyclicality
exponential rolling
temporal markers

Spatial View

Connectivity  Topology
regional aggregation
topology integration

Combined View

Efficiency  Flexibility
intrinsic Spatio-
temporal Dependencies

Our Method

Visual Space



Vision
Reconstruction

*Flexible and promising
for using vision models...*

Prediction